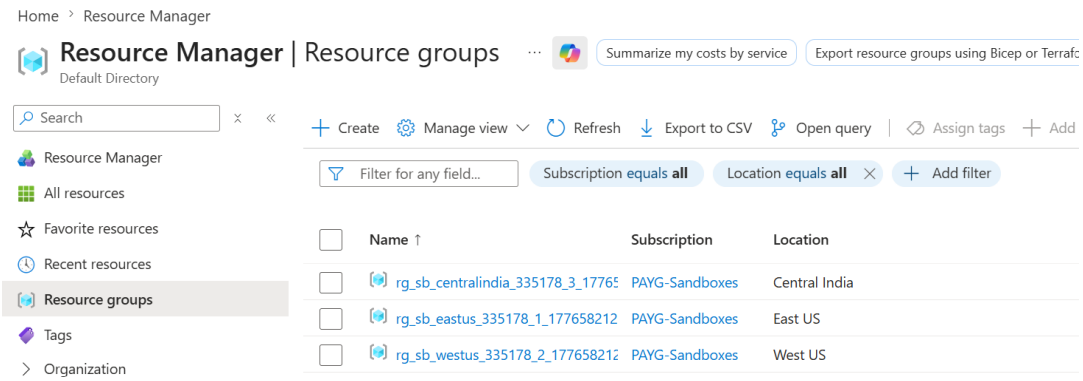
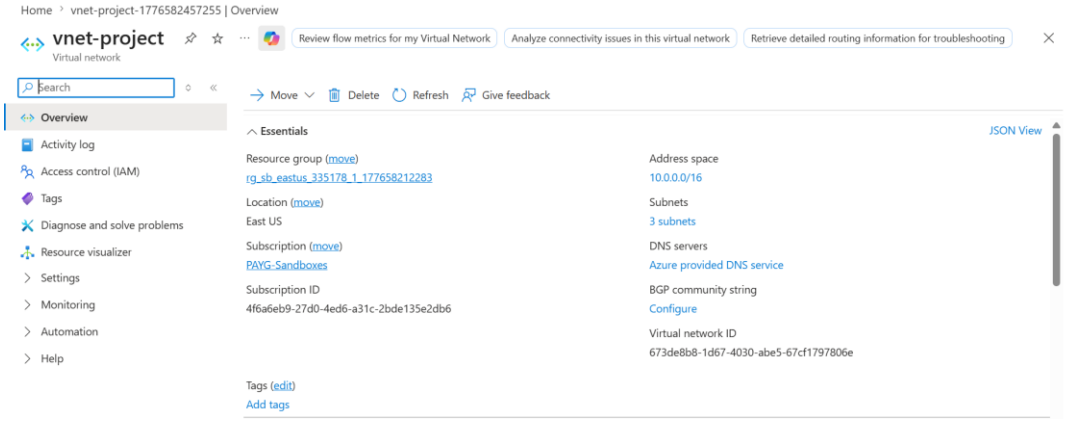
Project Requirements

1. Resource Organization



The resource group was pre-provisioned in the Whizlabs sandbox, so I used it to organize all deployed resources.

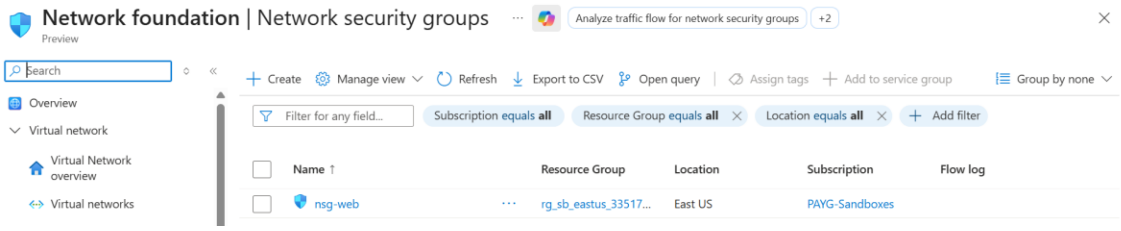
2. Networking Layer

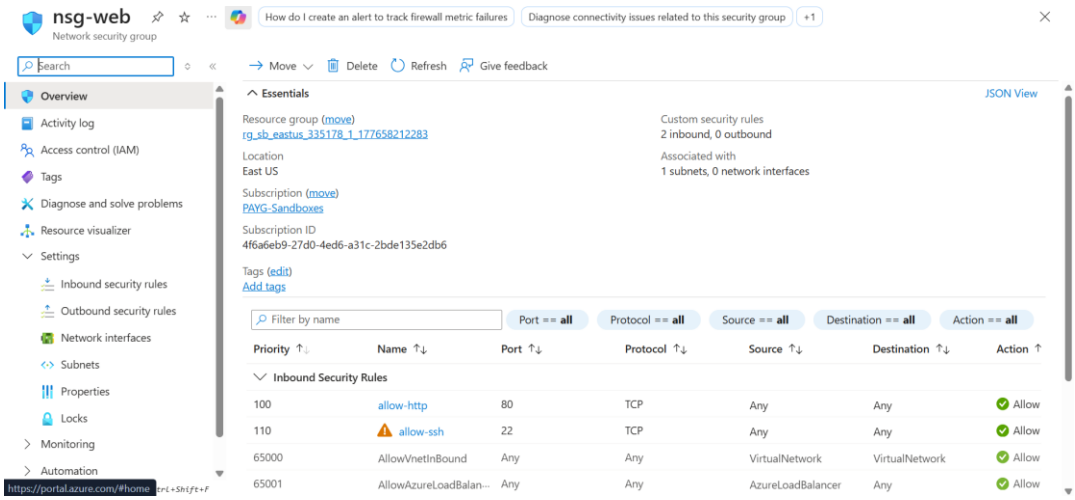


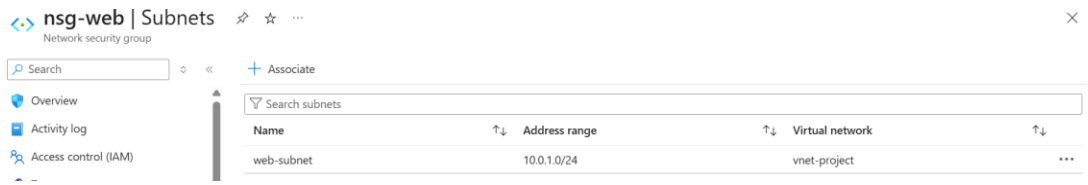
built a **3-tier architecture**:

- Web → frontend (VMs)
- App → backend (future use)
- DB → database layer

- Network Security Groups (NSGs)

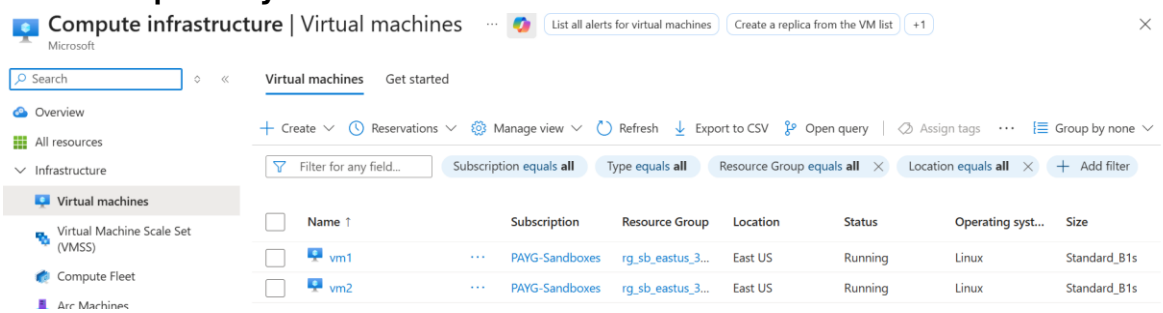






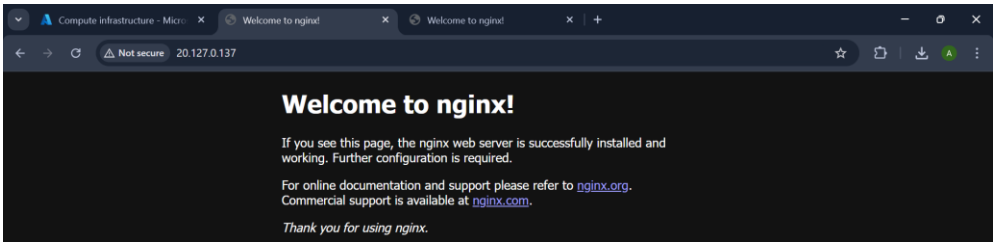
I implemented a Network Security Group to control inbound traffic, allowing only HTTP and SSH access to the web tier.

3. Compute Layer



Vm1

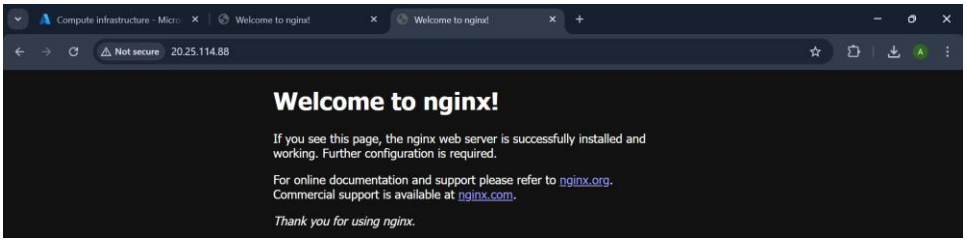
ssh ahlam@20.127.0.137
sudo apt update
sudo apt install nginx -y
sudo systemctl start nginx



sudo nano /var/www/html/index.nginx-debian.html

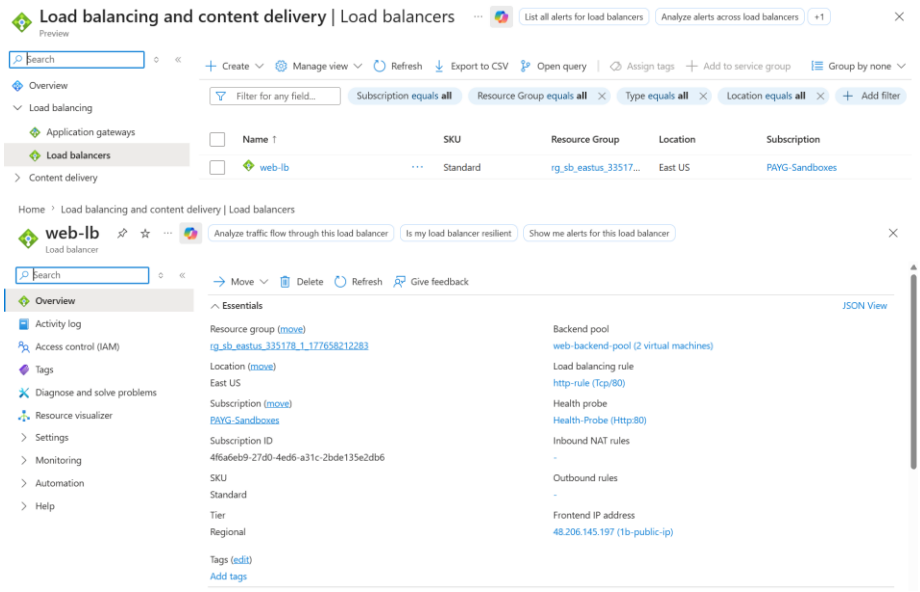
Vm2

ssh ahlam@20.25.114.88
sudo apt update
sudo apt install nginx -y
sudo systemctl start nginx

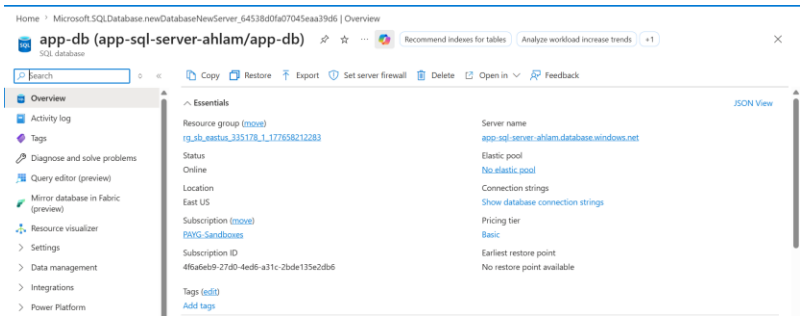


sudo nano /var/www/html/index.nginx-debian.html

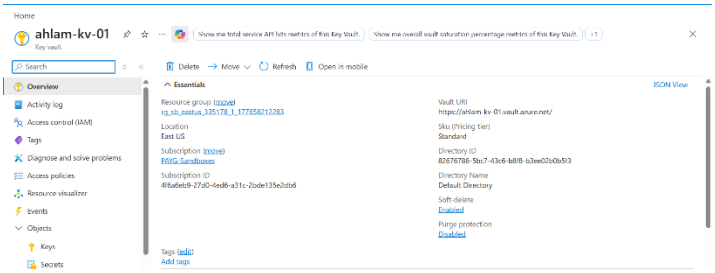
4. Load Balancing

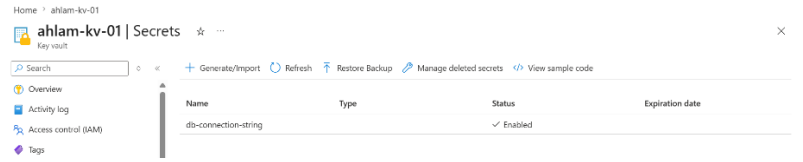
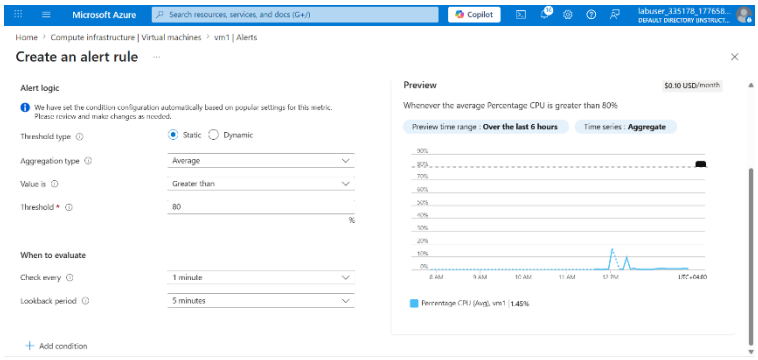
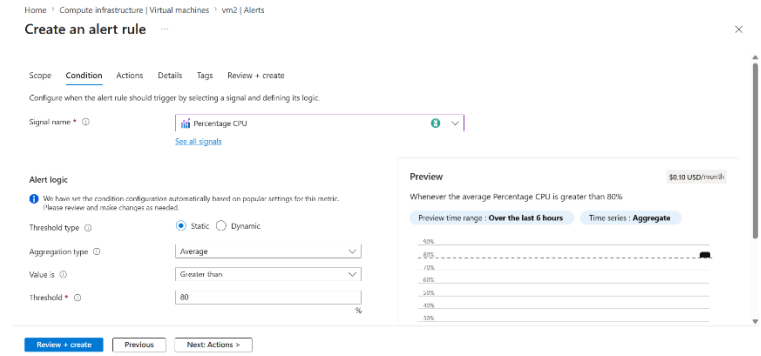


5. Data Layer



6. Security





7. Monitoring & Logging